Maddalena **Torricelli**



Digital Hub

maddalena.torricelli

maddalenatorricelli.me

LinkedIn

Google Scholar

GitHub

ORCID

Languages

IT (Native)

EN (Advanced)

ES, FR, DE (Basic)

Code Stack

Python (Pandas, NumPy, SciPy, NetworkX, Gensim, Matplotlib, Seaborn, PyTorch)

R (ggplot2, caret, glmnet)

HTML, CSS, JavaScript (basic knowledge)

LaTeX

DataViz (d3.js, Vega Lite, Google Data Studio)

Intensive course on effective visualization of data with d3.js and Vega Lite (Big Dive, Torino (IT), 2020)

WORKING EXPERIENCE

Postdoctoral Researcher — 2025 --
Network Science Institute, Northeastern University -- Boston (USA)

Research topics: network science and data analytics applied to sports, with primary focus on football; leveraging complex systems methods for tactical analysis, team dynamics, and player performance.

Visiting research period at ETH Zurich, in collaboration with Prof. Ulrik Brandes at the Social Networks Lab, focusing on a football analytics project (May 2026)

Events organization: Hackathon in collaboration with Sports Analytics Club at Northeastern University and PySport, Complexity72h (2026), Complexity and Sports satellite at CCS 2026

Postdoctoral Researcher — 2021 -- 2023
City St. George's, University of London -- London (UK)

Research topics: polarization and misinformation spread on social media about climate change; quantifying social perception of climate change and natural disasters.

Teaching activity: Number Theory and Cryptography - Mathematics Dept.

Events organization: Data Natives (2022, 2023), Complexity72h (2023)

EDUCATION

PhD in Data Science and Computation — Defense: March 21st, 2022
Alma Mater Studiorum Università di Bologna, Italy

Thesis: Embedded representations of social interactions

MSc in Physics of Complex Systems — Graduation: October 24th, 2017
Università degli Studi di Torino, Italy

Thesis: Party cohesiveness in the Italian Parliament (1987-2013)

BSc in Physics and Astrophysics — Graduation: June 26th, 2015
Università degli Studi di Firenze, Italy

Thesis: Collective motion as a critical phenomenon

SELECTED PUBLICATIONS

Evolution of spatial structure, passing network patterns, and gameplay intensity in elite women's and men's football (2020--2025) — May 2026
Scientific Reports

Paper resulting from Complexity72h (2025). Presented at ASI Summit, 31.01.2026, Houston, TX (USA)

Unveiling the drivers of active participation in social media discourse Feb 2025
Scientific Reports
Paper resulting from Complexity72h (2023).

Temporal dynamics of goal scoring in soccer Jan 2025
arXiv preprint

The role of interface design on prompt-mediated creativity in Generative AI May 2024
WEBSCI '24: Proceedings of the 16th ACM Web Science Conference
Presented at WEBSCI Conference, 23.05.2024, Stuttgart (DE)

How does extreme weather impact the climate change discourse? Insights from the Twitter discussion on hurricanes Nov 2023
PLOS Climate
Presented at CCS Conference, 20.10.2023, Salvador, Bahia (BR)

weg2vec: Event embedding for temporal networks Apr 2020
Scientific Reports
Presented at NetSci Conference, 29.05.2019, Burlington, VT (USA)

A full list of publications is available on my Google Scholar profile.

SELECTED TALKS

Hudl Performance Insights 2026 November 13th, 2026
Emirates Stadium, London
Research Stage talk: "The score shapes the team: Correlation functions of off-ball structure in football"

Meet a Scientist: Soccer Science Series July 5th, 2026
Museum of Science, Boston
It's all connected: what can networks reveal about soccer?

American Soccer Insights January 31st, 2026
Rice University, Houston, TX, USA
Harder, Shorter, Sharper, Forward: A Comparison of Women's and Men's Elite Football Gameplay

WEBSCI '24 May 23rd, 2024
Stuttgart, DE
The Role of Interface Design on Prompt-mediated Creativity in Generative AI

Let's talk polarization March 1st, 2023
Spotify
Navigating polarization in online climate discourse

IN THE MEDIA

The Future of Soccer Is Being Studied in Boston Oct 2026
La Stampa USA
Interview on network science in football and Italy's national team

Can Network Science Predict the World Cup? Jul 2026
Network Science Institute, Boston
Interview for Northeastern University about the 2026 FIFA World Cup

Argentina-Cape Verde preview: It's a matchup between Messi vs. Vozinha Jul 2026
Northeastern Global News
Match analyses for the 2026 FIFA World Cup

In treating soccer like a language, these researchers can predict the next word — Jan 2026
Northeastern Global News
Feature on the NetSI Sport research group

SCIENTIFIC LEADERSHIP & COLLABORATIONS

Complexity72h — Various editions
Complexity Next Gen
Participation in Complexity72h as co-organizer and tutor across multiple editions, supporting collaborative research projects that resulted in two scientific publications.

Complexity and Sports satellite at CCS 2026 — Oct 2026
Conference on Complex Systems
Organization of a satellite bringing together researchers working at the intersection of complexity science and sports analytics

"The Complexity of Sports" — 2026 --
npj Complexity Collection
Guest Editor of a collection on the application of complexity science to sports

Sónar — Jun 2023
BSC - CNS (Barcelona Supercomputing Center)
Collaboration with the EXPLAIN Project: analyzing lyrics and music of Sónar festival artists to detect patterns and track their evolution

Peer Review Activity — 2022 --
Journals & Conferences
Peer reviewer for Network Science, PLOS ONE, the Conference on Complex Systems (CCS), and the Network Science Society Conference